

# WATER LEVEL DETECTOR CIRCUIT

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MNLIMT001

EEE1006F project

## 1.Introduction

Through the knowledge gained from the EEE1006F module you were tasked with designing, building and delivering of a project to a client. The project involved establishing a water level detector system (WLDS) that takes into consideration the loadshedding conditions in South Africa.

## 2.Design

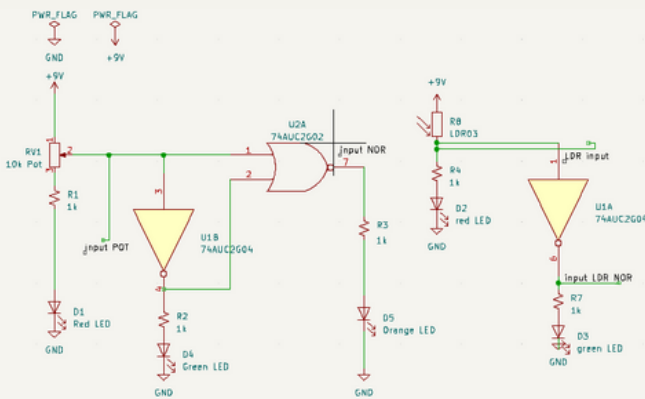


Figure 1: Schematic of digital circuit

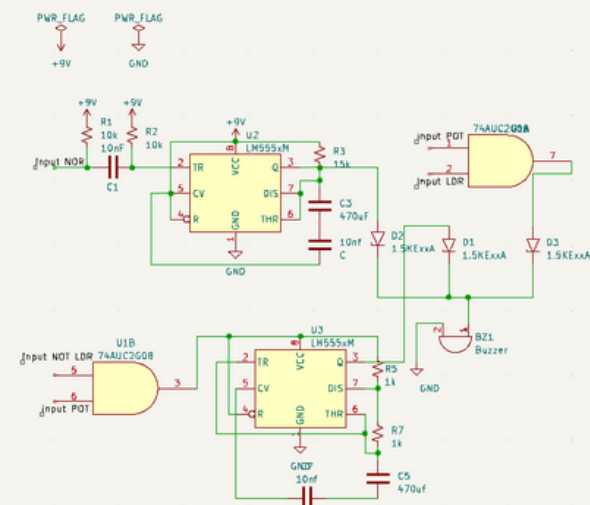


Figure 2: Schematic of warning circuit

## 4.Data Analysis

Circuit 1 consists of three digital outputs:  
HIGH -when the tank is full.  
MED-when the tank is half full.  
LOW -when the tank is empty.

The water level will be stimulated using a variable resistor to get the above digital inputs.

When the tank is full the resistance of the variable resistor is the greatest.

When the tank in empty the resistance of the variable resistor is the lowest.

When the tank is not full nor empty the variable resistor is between the 1st and last extreme and the output is sent to a NOR gate to output a HIGH output .

The loadshedding aspect of the circuit will be simulated through an LDR/switch. When the LDR is in light, the green LED will light up indicating that there is currently no loadshedding. If the LDR is in the dark, the red LED will light up indicating there is currently loadshedding.

The loadshedding circuit makes use of a NOT gate to get a HIGH output when undergoing loadshedding. (When the push button switch is pressed the red LED will switch on indicating that there is currently loadshedding, otherwise the green LED will switch on indicating there is currently no loadshedding.)

Circuit 2, the warning circuit has six different outputs. The outputs are as follows:

1. Light a green LED - when the tank is full with no loadshedding.
2. Light a red LED - when the tank is full during loadshedding
- 3.Buzzer beep once and light a green LED - when the tank is half full with no loadshedding.
- 4.Buzzer beep one and light a red LED - when the tank is half full with loadshedding.
- 5.Buzzer on and light a green LED - when the tank is empty with no loadshedding.
- 6.Buzzer beeping and light a red LED - when the tank is empty with loadshedding.

To get output 1 circuit 1's variable resistor must be the greatest. To get output 2 make use the red LED in the loadshedding circuit. To get output 3 the circuit makes use of a monostable 555 timer circuit.To get output 4 the circuit makes use of a monostable 555 timer circuit and the loadshedding circuit. To get output 5 and 6 an astable 555 timer was used as well as the loadshedding circuit for output 6.

## 4. Conclusion

The circuit works better in its separate parts than as a whole once everything is connected however all six outputs work as expected.

The digital inputs of the circuit overlapped with each other when turning the variable resistor between extremes. This resulted in a very short period for the MED output to work, this is as when the wiper was not at a specific part between extremes the medium output would not work. The monostable circuit output of the buzzer stayed on longer than the expected calculated value. The loadshedding circuit did not perform as well as expected as the green and red LED was minorly on when it was not supposed to be. The LDR needed direct light in order to output the expected outputs during no loadshedding. To improve this project the circuit should be soldered onto another board to prevent the components from being easily detachable. The neatness of the circuit can be improved by cutting wires to the correct length and by colour coordinating different wires for different functions. The circuit could also be improved by reducing the number of components and loading to create a more efficient and economical circuit.

## 5. References

[1] R. Keim, "Transimpedance amplifier: Op-amp-based current-to-voltage signal converter - video tutorial," All About Circuits, <https://www.allaboutcircuits.com/video-tutorials/op-amp-applications-current-to-voltage-converter/> (accessed May 16, 2023).

[2] W. Storr, "Pull-up resistor and pull-down resistor explained," Basic Electronics Tutorials, <https://www.electronics-tutorials.ws/logic/pull-up-resistor.html> (accessed May 19, 2023).

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